

AN2DL - Second Challenge Report

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1 Introduction

Regarding this four-class image classification problem, we employ a patch-based approach where images are subdivided into smaller regions. These patches are processed using **Multiple Instance Learning (MIL)** aggregation techniques to produce a slide-level prediction.

2 Problem Analysis

The dataset comprises 691 pathology images coupled with pixel-level annotation masks. The samples are categorized into four molecular subtypes: **Luminal A (30%)**, **Luminal B (32%)**, **HER2(+)** (27%), and **Triple Negative (11%)**. As indicated by these statistics, the dataset exhibits significant **class imbalance**, with **Luminal B** being the most prevalent and **Triple Negative** being underrepresented.

For feature extraction, we extract patches from annotated regions to form MIL bags (see Figures 1 and 2). However, limited mask coverage in some images results in sparse bags, which risks overfitting.

To address this, we expanded our sampling criteria:

1. **Boundary Inclusion:** We extract patches from the mask periphery to increase bag size and capture contextual information.

2. **Normal Tissue Padding:** If the patch count remains insufficient, we supplement the bag with normal tissue patches. This ensures consistent bag sizes and helps the model distinguish tumor features from the background.

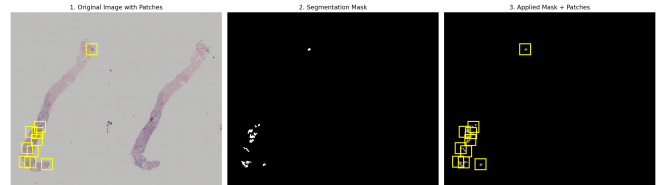


Figure 1: Example of extracted patches from a pathology image (0000)

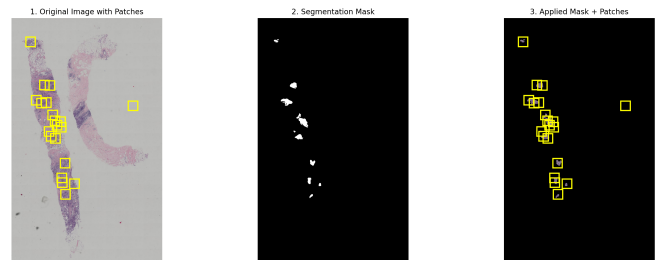


Figure 2: Example of extracted patches from a pathology image (0002)

3 Method

3.1 Noisy image recognition

We observed that some images in the dataset are noisy or corrupted (Orcs and Slime), which can negatively impact model performance. To address this, we used two layers of defense:

- **Color Check:** Histology images are never green. If an image has significant green pixels (common for orcs/slimes), it is immediately trashed.
- **CLIP Classification:** Use `openai/clip-vit-large-patch14` to identify and filter out these noisy images from the training set. We create text prompts like "a histology image" and "a green slime histology image" to classify images. If an image is classified as "green slime histology image" with high confidence, it is marked as noisy.

Finally, we removed **101** noisy images from the training set, get **590** clean images for training.

3.2 Data Augmentation

We utilized the **ImageNet AutoAugment policy** to align with our backbone’s pre-training on the ImageNet dataset. This facilitates effective transfer learning by ensuring input variations remain compatible with the pre-trained weights, while introducing diverse regularization (e.g., geometric and color shifts) to prevent overfitting on the limited pathology data.

3.3 Dual-stream MIL Network

We employ a Dual-Stream Multiple Instance Learning (MIL) architecture [1] to synergize local morphological details with global spatial context. As illustrated in Figure 3, the framework operates via two parallel pathways: a *Local Stream* (Bag of Instances) and a *Global Stream* (Contextual View), which are fused for final classification.

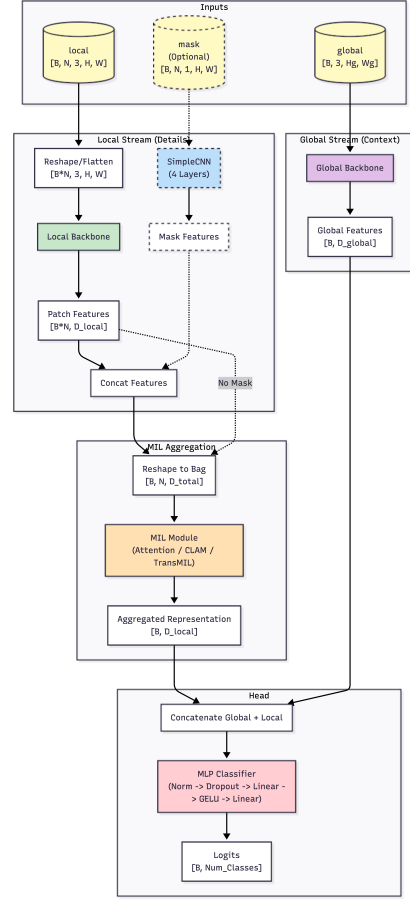


Figure 3: Dual-Stream MIL Architecture

Local Stream This stream processes the bag of extracted patches. A shared backbone extracts feature embeddings from each instance, which are subsequently projected into a lower-dimensional space via a bottleneck layer.

Optional Mask Fusion To leverage the available segmentation annotations, we integrate an optional mask fusion step. Binary mask information is concatenated with patch embeddings to explicitly guide the network toward relevant pathological regions.

MIL Aggregation We apply a permutation-invariant aggregation function to the instance embeddings. This synthesizes the varying number of patch features into a fixed-length, slide-level representation.

Global Stream To capture macro-architectural patterns often lost during tiling, the global stream processes a downsampled view of the entire image

using a backbone. Similar to the local stream, these global features are compressed via a bottleneck projection before fusion.

4 Experiments

We conducted extensive experiments to evaluate the Dual-Stream MIL model, assessing the impact of patch selection, aggregation strategies, and backbones.

Patch Extraction Patches of 224×224 pixels were extracted with a stride of 112. Using the binary masks, we discarded patches containing less than 50% tissue. To handle class imbalance during training, we utilized a **WeightedRandomSampler**. Empirically, we determined that a bag size of approximately 8 patches offered the optimal trade-off between computational efficiency and representational power.

Aggregation Method We benchmarked various MIL aggregation techniques, including Max Pooling, Mean Pooling, Attention-based Pooling, Gated Attention, TransMIL, and **CLAM** (Clustering-constrained Attention Multiple instance learning). We found that **CLAM** yielded the best performance, effectively learning instance-level importance to refine slide-level predictions.

Optimizer Training was performed using the **AdamW** optimizer with a learning rate of 1×10^{-4} and weight decay of 1×10^{-2} . We also explored the **Lion** optimizer but observed more stable convergence with AdamW.

Table 1: Experimental measurements for different models.

Model(local + global)	F1-score in validation
resnet18+resnet18	0.3674
efficientnet_b0+efficientnet_b0	0.3734
densenet121+resnet18	0.3828
resnet18+convnext_tiny	0.3482
convnext_tiny+resnet18	0.4226
resnet50+resnet18	0.3116

5 Discussion

Our experiments demonstrate that the architectural choice for the Local and Global streams significantly impacts performance. As shown in Table 1, the combination of **ConvNeXt-Tiny** (Local) and **ResNet-18** (Global) yielded the highest F1-score of **0.4226**, outperforming other configurations. For the model selection throughout the entire experimental process, please refer to the A.

We attribute this success to two main factors. First, ConvNeXt designs modernize the standard ResNet architecture with depth-wise convolutions and larger kernels, allowing it to capture more robust morphological features from local patches without a massive increase in parameters. Second, the use of a lightweight **ResNet-18** for the global stream proved sufficient for capturing macro-spatial context without overfitting.

Notably, larger backbones such as **ResNet-50** performed poorly (0.3116 F1-score). Given the limited size of our clean training set (590 images), deeper models likely suffered from severe overfitting, failing to generalize to the validation set. This confirms that for small-scale histology datasets, model compactness and inductive bias (as seen in ConvNeXt) are more critical than raw model depth.

6 Future Improvements

We propose the following strategies:

- **Contrastive Representation Learning:** Rather than relying solely on ImageNet weights, we propose using self-supervised contrastive frameworks (e.g., SimCLR or MoCo) on the pathology patches. This would allow the backbone to learn invariant tissue representations by pulling similar features together and pushing dissimilar ones apart before classification begins.
- **Generative Data Synthesis:** To overcome the limited dataset size, we could employ conditional Generative Adversarial Networks (cGANs) or Diffusion models conditioned on the segmentation masks. This would allow us to "synthesize" new, realistic training images with diverse textures while preserving the original morphological structures.

References

- [1] B. Li, Y. Li, and K. W. Eliceiri. Dual-stream multiple instance learning network for whole

slide image classification with self-supervised contrastive learning. In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 14313–14323. IEEE.

A Appendix: Full Process Experimental Results

Table 2: Starting from a **simple CNN model**, we use the resized images as input directly.

Model	F1-score in validation	Note
Simple CNN (4 layers)	0.2759	Simple CNN is hard to learn since limited dataset.
ResNet18	0.3090	Severe underfitting, loss oscillates severely.
ResNet50	0.3250	Similar to ResNet18 but slightly better.
EfficientNet_b0	0.3014	Less parameters, faster but still bad fit
ConvNeXt_tiny	0.3322	Severe overfitting, but the effect has improved, consider work on data

Table 3: Use **multi-instance learning (MIL)** with extracted patches as input.

Model	patch size * number	aggregation	F1-score in validation	Note
ConvNeXt_tiny	64*8	Max Pooling	0.3412	Simple MIL, better than using whole image
ConvNeXt_tiny	64*16	attention	0.3630	Larger patch size improves performance
ConvNeXt_tiny	224*8	clam	0.3907	No resize for model input, 224 could be better
ConvNeXt_tiny	224*16	—	—	CUDA out of memory

Table 4: Use dual stream, put the mask into the local stream.

Model(local + global)	F1-score in validation	Note
convnext_tiny+resnet18	0.4623	Get the highest score, but normal (0.4008) in submission
convnext_tiny+convnext_tiny	0.4319	Severe overfitting! Global stream is too strong
resnet18+resnet18	0.3674	Loss extreme volatility, model is hard to learn